

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. BPT-502: Classical Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

- (a) Define non-holonomic constraints with examples. Show that the constraints of a rigid body are of holonomic type.
- (b) The Lagrangian of a system with N degrees of freedom is given by:

$$L(q_i, \dot{q}_i) = \frac{1}{2} \sum_{i=1}^N m_i \dot{q}_i^2 - \sum_{i,j=1}^N k_{ij} (q_i - q_j)^2$$

Find the equations of motion for the system, where $i = 1, 2, \dots, N$.

- (c) The Hamiltonian of a system is given by:

$$H(q, p) = \frac{p^2}{2\alpha} + \frac{kq^2}{2} e^{-\alpha t} - \beta q p e^{-\alpha t}$$

where α, β, k are constants. Find the Lagrangian of the system.

- (d) Find five different vector potentials $\vec{A}$ which produce the same uniform magnetic field $\vec{B} = 2\hat{i} + 3\hat{j} + 5\hat{k}$.
- (e) Find the central force that will cause a point particle of mass m to describe a circular orbit passing through the center of force.

Question 2.

- (a) A particle of mass m is constrained to move on the surface of a cylinder $x^2 + y^2 = R^2$. The particle is subjected to a force directed towards the origin and proportional to the distance of the particle from the origin. Construct the Lagrangian of the system. [7]
- (b) What is meant by *homogeneity of space*? Derive the conservation law that follows from it. [7]

Question 3.

- (a) A particle of mass m is sliding on a smooth wire whose equation in cylindrical coordinates is given by $\rho = (z + a\theta)^2$, $z = b$, where a, b are constants. Find the equation of motion for the particle using the Lagrange multiplier method. [8]

- (b) Write down the Lagrangian of a charged particle in an electromagnetic field. Construct the Hamiltonian for the system and hence obtain the canonical equations of motion. [6]

OR

- (a) A particle of mass m is moving on a plane under an attractive central force given by:

$$F(r) = -\frac{\alpha}{r^2} - \frac{\beta}{r^3}, \quad \alpha, \beta > 0$$

- (i) Find the effective potential for this system and plot it approximately. [4]
 (ii) Discuss the motion of the particle qualitatively for various total energy values of the particle. [4]
 (iii) Show that the angular momentum of the particle is conserved. [3]
 (iv) Show that the position vector of the particle sweeps out equal areas in equal times. [3]

Question 4.

- (a) Define canonical transformation. Construct all possible generating functions (F_1, F_2, F_3, F_4) for the canonical transformation:

$$Q = \frac{1}{p}, \quad P = qp^2$$

[7]

- (b) Calculate the Poisson bracket $\{v_i, v_j\}$, where v_i and v_j represent the i -th and j -th components of velocity of a charged particle in the magnetic field $\vec{B}$. [7]

OR

- (a) Define normal coordinates and normal frequencies of an oscillating system. Find the normal modes in the vibration of a CO_2 molecule, by considering a linear model of three masses $O - C - O$. What is the physical meaning of zero normal frequency? [9]

- (b) A particle of unit mass moves under a one-dimensional potential:

$$V(x) = x^4 - 2x^2 + 7$$

Find the frequencies of small oscillations about the stable equilibrium positions of the mass. [5]

Question 5.

- (a) In a dynamical system with two degrees of freedom, the Lagrangian is given by:

$$L = \frac{\dot{q}_1^2 + \dot{q}_2^2}{2(a + bq_2)} - (cq_1^2 + dq_2^2)$$

where a, b, c, d are constants. Deduce the existence of a constant of motion. [6]

- (b) Consider a simple harmonic oscillator in one dimension with angular frequency ω . Show that the quantity:

$$u(q, p, t) = \ln(p + im\omega q) - i\omega t$$

is a constant of motion.

[6]

- (c) Find a suitable scaling under which the dynamics of the system described by:

$$L(\dot{q}_1, \dot{q}_2, q_1, q_2) = \dot{q}_1^2 + \dot{q}_2^2 + 5\dot{q}_1\dot{q}_2 - 3q_1^2 - 5q_2^2 - 11q_1q_2$$

remains invariant.

[8]